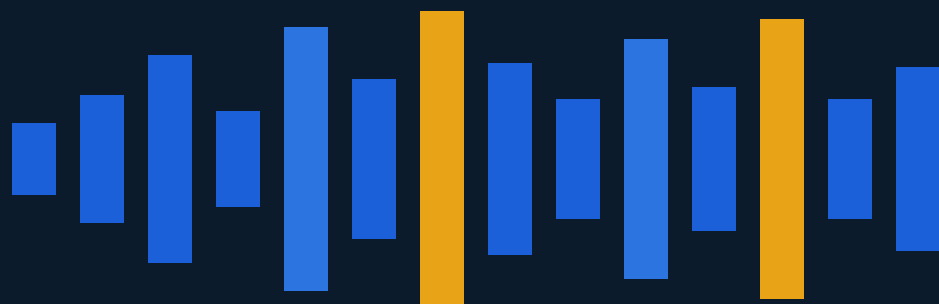


VANI

Speak in one Indian language. The other phone hears it in hers. No network, no cloud, nothing leaves the device.



10 Indic languages

Zero cloud calls

Text-only link

PROBLEM STATEMENT ID

SIH26173

PROBLEM STATEMENT TITLE

iTantra - Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links

THEME

Smart Automation

PS CATEGORY

Software

ORGANISATION

Indian Space Research Organisation (ISRO)

TEAM NAME

VANI

TEAM ID

fill from portal

Every figure in this deck is measured from the running app on an Android device. Nothing is projected or illustrative.

Say it in Hindi. It arrives spoken in Marathi.

VANI bundles speech recognition, neural translation and speech synthesis inside both phones. Only the recognised sentence crosses the link - about 90 bytes of text where the same utterance would cost roughly 91 KB of raw audio. The receiving phone rebuilds it as audible speech in the listener's own language.

THE GAP THE PROBLEM STATEMENT NAMES

Voice costs bandwidth. 16 kHz mono PCM is about 32 KB for every second of speech.

Text assumes literacy. A typed message excludes the people voice is meant to reach.

The two sides differ. Caller and responder often share no written language at all.

WHAT VANI DOES ABOUT IT

Tap and speak. Hold the mic, release it, and the utterance is captured on the phone.

Recognised and translated on-device. IndicConformer, then IndicTrans2 - Hindi to Marathi, never through English.

Spoken on the far phone. IndicF5 renders it as a replayable voice note in that language.

NOVELTY 1

No cloud, no token, no cost

Every stage runs in-process. No network call; works in airplane mode.

NOVELTY 2

Indic to Indic, no English pivot

Most Indian stacks route through English and lose the idiom.

NOVELTY 3

A different voice per language

IndicF5 is zero-shot prompted, so Marathi is not read in a Hindi voice.

Three neural models per phone. The link carries text.

SENDER - ANDROID PHONE, OFFLINE

MicRecorder

AudioRecord - 16 kHz mono PCM16

↓ log-mel 80 x T

ConformerStt

IndicConformer-HI - CTC greedy - INT8

↓ Devanagari text

TranslatorEngine

IndicTrans2 320M - [hi, mr] token prefix

UTF-8



THE LINK

~90 B

of text crosses the link

against ~91 KB of raw 16 kHz PCM for the same sentence.

Bluetooth RFCOMM / Wi-Fi Direct,
the transport SIH26173 names.

Today: in-app loopback.

UTF-8



RECEIVER - ANDROID PHONE, OFFLINE

DhVaaniEngine

IndicF5 - flow matching - 8 Euler steps

↓ conditioned on

Reference prompt

per-language voice clip - 320 mel frames

↓ 24 kHz PCM16

Player

AudioTrack - one buffer per utterance

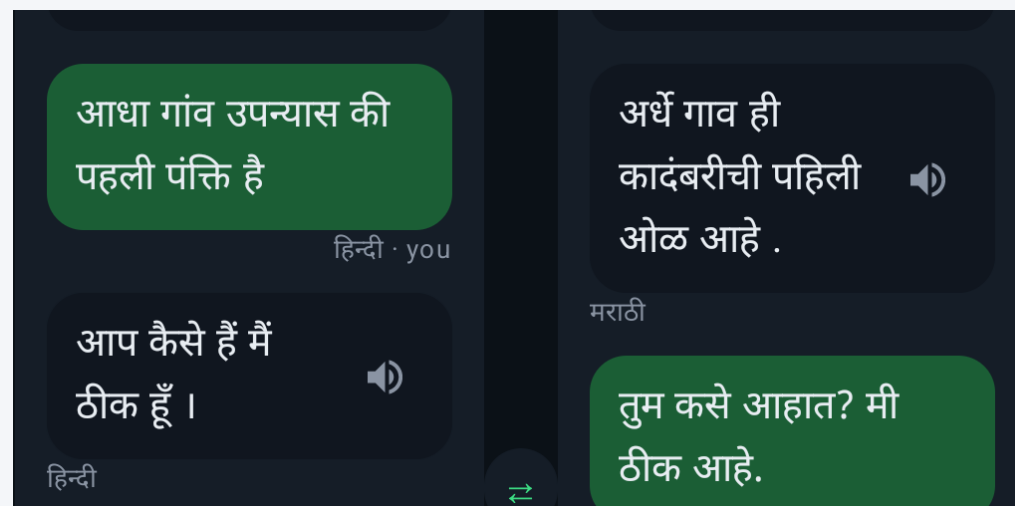
HOW IT IS BUILT

Kotlin 1.8, Jetpack Compose Material3, 3,094 lines, no XML. One Activity, two panels, no navigation graph.

ONNX Runtime Android 1.22.0 on the CPU execution provider. Threads = cores/2 capped at 4; more measured slower.

5 of 6 graphs INT8-quantised. Mel, FFT and iSTFT are Kotlin - no NDK, no native audio code.

STT 137.3 MB - MT 528.5 MB - TTS 185.3 MB - voices 6.8 MB. 857.9 MB of assets streams from a 718.9 MB install.



Captured from the built app: Hindi in, Marathi out, replay cached.

Measured on the running app, walls included.

13.1%

word error rate

4.8% character error rate

Scored on-device against AI4Bharat Kathbath Hindi: 40 clips, 207 s of speech, 19 speakers.

0.23x

faster than real time

627 ms

Hindi to Marathi

12.9 s

mic release to first audio

WHERE THE 16.2 SECONDS GO



STT 1,092 ms - MT 627 ms - **TTS render 11,199 ms** - playback 3,307 ms

The flow-matching solve is 69% of the path. Sweeping steps, guidance and threads changed nothing.

Android emulator, debug build, single run per figure. Cold model load adds 52.9 s.

RISKS AND HOW WE CLOSE THEM

TTS render dominates the end-to-end path.

8.5-11.2 s to make 2.2-3.1 s of audio; steps, prompt length and threads were swept and none moved it.

Speech recognition covers Hindi today.

One IndicConformer export is bundled; the family ships per-language models, so this is an asset swap.

The link is in-app loopback, not radio yet.

Every stage either side of the transport is built; RFCOMM pairing and a 90-byte text frame are what is left.

719 MB install, 858 MB of models.

Already inside the 800 MB ceiling without cutting quality; INT4 weights are the next reduction.

Cost and deployment: zero inference cost per message, no server estate, no API quota. One APK, installed once, works in airplane mode.

Voice identity: 7 of 19 reference clips pass the quality gates, so 7 languages borrow a donor voice; 11 screened replacements are ready.

A thousand times less to send. And nobody has to read.

~1,000x

smaller payload on the constrained link

2.9 s of speech = 92,800 bytes at 16 kHz mono PCM16.

Its 7-word Hindi transcript = 90 bytes of UTF-8. Same meaning, 0.1% of the airtime.

Low-literacy users

Neither side has to read or type.

Injured or panicking

A mic hold is one gesture.
Composing text is not available.

Cross-language teams

Responder and caller rarely share
a written language.

Narrow links

90 bytes fits where 91 KB of audio
cannot go at all.

AGAINST THE SIH26173 MINIMUM DEMO

BUILT

Speak a sentence in one Indian language into one phone.

BUILT

Show it transcribed on-device - 13.1% word error rate, 4.8% character error rate, measured.

BUILT

A second phone speaks it back intelligibly in its own language, as a replayable voice note.

OPEN

Carry the text over Bluetooth rather than the in-app loopback - the one remaining integration step.

BUILT

Report end-to-end latency and word error rate - 12.9 s and 13.1%, both from the running app.

Nothing on this slide is projected. Every figure came from a run of the installed build.

What we built on, and what we measured ourselves.

BUNDLED MODELS - RUN INSIDE THE APK

ai4bharat/indicconformer_stt_hi_hybrid_ctc_rnnt_large

Speech recognition. CTC branch, INT8 ONNX export, 137.3 MB.

ai4bharat/indictrans2-indic-indic-dist-320M

Machine translation, Indic to Indic, no English pivot. 528.5 MB.

ai4bharat/IndicF5

Zero-shot flow-matching synthesis, 8 Euler steps, CFG 2.0. 185.3 MB.

PEER-REVIEWED BASIS

[arXiv:2305.16307](#)

IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages

[arXiv:2409.05356](#)

IndicVoices-R: Unlocking a Massive Multilingual Multi-speaker Speech Corpus for Scaling Indian Speech

DATA AND LICENCE

ai4bharat/Kathbath - CC-BY-4.0

Source of the 22 reference voice clips shipped in the app, and of the 40-clip Hindi set the word error rate is scored on. Each clip's exact transcript is bundled beside it and used verbatim.

Clips are screened by measured pitch, internal-pause fraction and token density before shipping.

RUNTIME AND TOOLING

Microsoft ONNX Runtime 1.22.0 (onnxruntime-android), CPU execution provider. No NNAPI or GPU delegate is used.

Kotlin 1.8.22, Jetpack Compose Material3, AGP 7.4.2. minSdk 26, target 34, arm64-v8a and x86_64.

models.ai4bharat.org - model cards for all three bundled networks.

Measured by this team, not cited: word and character error rate over 40 Kathbath Hindi clips scored with normalised Levenshtein; per-stage latency from on-device logs; and the Euler-step, guidance and thread sweeps behind the feasibility slide.